PART XIII

Android Platform

24. Why is Android considered a software platform?

Because it provides a complete stack including an operating system, middleware, runtime environment, libraries, and application framework that allows developers to build and run software applications across diverse hardware.

25. What is AOSP?

AOSP stands for Android Open Source Project, the open-source software stack maintained by Google that serves as the foundation for Android OS builds.

26. What is an API level?

An integer value that uniquely identifies the framework API revision provided by a specific version of the Android platform.

27. What is the purpose of the Android Runtime?

It executes Android application bytecode (DEX format) and manages memory, garbage collection, and runtime performance optimization using ART.

28. What is the purpose of a HAL?

The Hardware Abstraction Layer (HAL) provides standard interfaces that expose device hardware capabilities, such as cameras, audio, and sensors, to higher-level Android framework components.

29. Why is the Linux kernel important to Android?

It handles core operating-system functions such as process management, memory allocation, security, power management, and hardware drivers.

30. What services does the Android framework provide?

It provides high-level APIs for UI components, resource management, activity management, notifications, location services, content providers, and security permissions.

31. Why must Android applications account for different device configurations?

Because Android runs on a wide variety of hardware with different screen sizes, orientations, resolutions, chipsets, sensors, and system locales.

Mobile Device Features

32. What are basic mobile-device features?

Core capabilities such as cellular/Wi-Fi connectivity, display/touchscreen input, speakers, microphones, and battery management.

33. What are advanced mobile-device features?

Specialized features such as biometric authentication, high-end GPUs, foldable displays, multi-lens cameras, NFC, and specialized sensors.

34. What is the difference between a hardware and software feature?

Hardware feature: A physical component built into the device, such as a camera, GPS chipset, or NFC reader.

Software feature: A capability provided by the operating system or application software, such as portrait-mode processing or face detection.

35. Why should developers distinguish required and optional features?

To ensure apps can be properly filtered in Google Play and gracefully degrade functionality on devices that lack non-essential hardware.

Connectivity

36. What is Wi-Fi?

A wireless networking technology that allows devices to connect to a local area network and access high-bandwidth internet over radio waves.

37. What is Bluetooth?

A short-range wireless technology designed for exchanging data between fixed and mobile devices over short distances.

38. What is BLE?

Bluetooth Low Energy (BLE) is a power-optimized variant of Bluetooth designed for low-power devices and short, periodic data transfers.

39. What is NFC?

Near Field Communication (NFC) is a short-range wireless communication technology that enables simple data exchange between devices brought within a few centimeters.

40. When would cellular connectivity be necessary?

When an application requires internet access while the user is outside the range of available Wi-Fi networks.

41. How should a developer select a connectivity technology?

Based on range, power consumption, data-transfer speed, bandwidth, proximity requirements, and hardware availability.

Sensors and Location

42. What is an accelerometer?

A sensor that measures acceleration along three physical axes (X, Y, and Z) to detect device movement, orientation, and tilt.

43. What is a gyroscope?

A sensor that measures the rate of rotation around the three physical axes.

44. What is a magnetometer?

A sensor that measures magnetic-field strength across three axes and is commonly used as a compass.

45. What are environmental sensors?

Sensors that measure ambient conditions such as light, pressure, temperature, and humidity.

46. What is GNSS/GPS?

Global Navigation Satellite System / Global Positioning System is a satellite-based navigation system used to determine a device's geographic location.

47. What is geofencing?
A feature that defines virtual geographic boundaries and triggers actions or notifications when a device enters or exits those boundaries.

48. Why is location considered sensitive information?
Because location data can reveal personal habits, home and work locations, routines, and movement patterns, creating privacy and safety risks.

Security and Privacy
49. What is application sandboxing?
A security mechanism that isolates each Android app within its own Linux process and user ID, preventing unauthorized access to other apps' private data.
50. What is SELinux?
Security-Enhanced Linux (SELinux) is a mandatory access-control architecture integrated into Android to enforce security policies and limit process privileges.

51. What is Verified Boot?
A process that cryptographically validates the integrity and authenticity of software during the boot sequence.

52. Why are Android permissions necessary?
To restrict application access to sensitive user data and hardware features such as contacts, location, cameras, and microphones unless authorized.

53. What is data minimization?
The practice of collecting, processing, and storing only the minimum personal data necessary for an application's purpose.

54. What is the difference between security and privacy?

Security: Protecting data and systems against unauthorized access, attacks, and corruption.
Privacy: Ensuring appropriate handling, consent, transparency, and user control over personal data.
55. Why should an application provide alternatives when a user denies permission?
To maintain a good user experience and allow non-dependent features to continue functioning without crashing.
Application Requirements
56. Why should feature selection begin with application requirements?
To ensure the application requests only necessary capabilities, reducing resource consumption, unnecessary permissions, and device incompatibility.
57. How do device capabilities affect application design?
They influence performance limits, UI layouts, online/offline strategies, and fallback mechanisms for unavailable hardware or sensors.

58. Why should developers consider battery consumption?
High battery consumption can result in poor user experiences, negative reviews, reduced performance, and system throttling.

59. Why should developers provide fallback mechanisms?
To ensure the application can continue operating when hardware, connectivity, or permissions are unavailable.

60. Why is compatibility testing important?
To verify that an application works reliably across different screen sizes, Android versions, manufacturers, and hardware configurations.

61. How should a developer determine whether a feature is essential?
Determine whether the application's core purpose can function without it. If it can, the feature should generally be optional.

SHORT QUIZ
Part A — Identification
62. Android Open Source Project (AOSP)
63. Android Runtime (ART)

64. Hardware Abstraction Layer (HAL)

65. Linux Kernel

66. Android Studio

67. Kotlin

68. Android Jetpack

69. Jetpack Compose

70. API Level

71. Google Play Store

Part B — Multiple Choice
11. B. Linux Kernel
12. C. Abstract hardware-specific implementations

13. B. ART

14. A. Jetpack Compose

15. C. Google

APPLIED ASSESSMENT SCENARIO
Scenario Context
A developer is creating a weather application that displays current weather, location-based weather, weather alerts, and forecast information.
72. Which connectivity technologies are required?

Wi-Fi or Cellular network connectivity is required to fetch dynamic weather data and alerts from remote APIs.

73. Is location required?
No. Location is optional. Users can manually search for and select cities without sharing their location.

74. Is precise location necessarily required?
No. Coarse location is generally sufficient for city-level weather forecasts.

75. Should background location be used?
Only if background severe-weather alerts or automated geofenced weather notifications are a core feature. Otherwise, location should be obtained only when the app is in the foreground.

76. Which permissions may be necessary?

INTERNET
ACCESS_NETWORK_STATE
ACCESS_COARSE_LOCATION
ACCESS_FINE_LOCATION — only if more precise location is needed
POST_NOTIFICATIONS — for notifications on supported Android versions
77. What privacy concerns exist?
Privacy concerns include continuously tracking the user's location, unnecessarily storing location history, or sharing location data with third-party advertising networks.
78. What should happen if the user denies location permission?
The app should provide an alternative such as a manually selected default location or city-search field.

79. What should happen if the device has no Internet connection?
The application should display cached or previously fetched weather information along with a clear indication that the information may be outdated.

80. What data should be stored locally?

Cached weather data
Last-known forecast information
Saved cities/locations
User preferences, such as °C or °F
Timestamp of the last update
81. Which device feature is essential and which features are optional?
Essential: Internet/network connectivity.

Optional: Location sensors (GPS/network location provider) and push-notification capabilities, depending on the app's features.